

Exploring the eco-evolutionary competition-colonization tradeoff model

Introduction

Intratumoral heterogeneity (ITH), the coexistence of multiple clones with distinct phenotypic properties, poses a significant challenge for cancer treatment, contributing to therapeutic resistance and treatment failure in cancer patients. Tumors harbor diverse populations of cancer cells that differ across key phenotypic traits such as proliferation, invasiveness, metabolic profiles, and immune evasiveness. But how this diversity arises and is maintained is poorly understood. Elucidating these fundamental mechanisms will be critical to devising novel strategies to promote tumor homogeneity for improved therapeutic response. To address this question, we turn to evolutionary community ecology—a field that studies how species interact via factors like competition, predation, and mutualism, and how these interactions shape the evolution of community structure and species diversity over time.

In this paper, we use mathematical modeling to show how ITH may be an emergent property of the tumor’s community structure. We draw on modern eco-evolutionary theory to show that hierarchical competition organized along life history trade-offs can maintain stable coexistence even when pairwise interactions among cancer clones may lead to competitive exclusion. We specifically explore the competition-colonization trade-off wherein a high intrinsic growth rate comes at a cost of lower competitive ability. **Discuss cancer implications here.** We provide theoretical metrics for detecting robust vs. fragile community structure in tumors and identifying keystone competitors that allow clonal diversity to persist. This framework provides a mechanistic basis for understanding how tumor communities can support multiple clones with distinct phenotypic properties.

With this understanding, we explore strategies to promote tumor homogeneity via the collapse of community structure. In addition to targeting keystone competitors and assessing its cascading effects on ITH, we will examine how modifications to the shape of the trade-off between competition and colonization impact ITH, and investigate how we can use our knowledge of tumor community structure to predict the best sequential therapies for treatment in the context of evolutionary double bind therapy (first hit homogenizes, second hit eradicates). Finally, we introduce the idea of “ecological cascade therapy”: drugging a more easily-targetable clone and allowing the resulting cascade to eradicate the desired (e.g., resistant) clone.

The model

Major simplifications: ignoring plasticity (e.g., cancer cells can readily switch their invasive phenotype epigenetically by regulating EMT (epithelial-to-mesenchymal transition) genes and instead assuming that these traits are fixed and heritable). Must also be careful with how we define clones (phenotypically

rather than genetically).

Below is an almost-classic competition colonization tradeoff model. Actually, it is closer in spirit to the competition-mortality tradeoff of Adler and Mosquera (2000). It also includes the smoothing proposed by the same authors: species lower in the hierarchy can still win against those higher, but with a reduced rate, and only if their traits are similar. The evolving trait z is the proliferation capacity that determines how well a species can grow in the absence of competition; in return, it will be less competitive (so low values of z correspond to good competitors and slow growers, while a high z means weak competitor but fast grower). The model reads

$$\frac{dN(z_i)}{dt} = N(z_i) \left(b(z_i) - \sum_{j=1}^S a(z_i, z_j) N(z_j) \right), \quad (1)$$

where $N(z_i)$ is the population density of individuals with phenotype z_i (here “species”, “clone”, and “phenotype” will be used interchangeably), $b(z_i)$ is their intrinsic growth rate, S is the number of species, and $a(z_i, z_j)$ is the competitive effect of an individual with phenotype z_j on another with phenotype z_i . The bracketed term above is the per capita growth rate $G(z_i)$ of phenotype z_i (also known as the fitness generating function or G -function):

$$G(z_i) = b(z_i) - \sum_{j=1}^S a(z_i, z_j) N(z_j). \quad (2)$$

What makes this into a competition-colonization (or competition-mortality) tradeoff model is that (i) $b(z)$ is a monotonically increasing function of z , and (ii) having a lower value of z means not experiencing much competition from those with higher z , but species with higher z do experience competition from species with low z values. This can be implemented, e.g., via the sigmoid function

$$Q(z) = \frac{\tanh(z) + 1}{2}, \quad (3)$$

going from 0 at $z \ll z'$ to 1 at $z \gg z'$, with σ controlling the width of the transition (Figure 1). The competition kernel $a(z, z')$ can then be written as

$$a(z, z') = Q\left(\frac{z - z'}{\sigma}\right). \quad (4)$$

We do not yet specify the form of $b(z)$, but we will require that $b(0) = 0$ and $b(z < 0) < 0$, so a trait value of $z = 0$ corresponds to no growth and the highest possible position in the hierarchy without having negative intrinsic growth and thus necessarily going extinct.

Evolutionary branching

From Eqn 1, we can derive when there will be opportunities for diversification under evolutionary dynamics via small mutational steps. Say we have a single resident species with trait z . Its nonzero equilibrium density is computed by setting the per capita growth rate of Eqn 2 to zero:

$$G(z) = 0 = b(z) - a(z, z) N^*(z) \implies N^*(z) = b(z)/a(z, z) = 2b(z), \quad (5)$$

where we used the fact that $a(z, z) = Q((z - z)/\sigma) = Q(0) = 1/2$ from Eqn 3. The invasion fitness $G(z_m, z)$ of a mutant phenotype z_m in the presence of a resident at z then reads

$$G(z_m, z) = b(z_m) - a(z_m, z_m) \underbrace{N(z_m)}_0 - a(z_m, z) N^*(z) = b(z_m) - 2b(z)a(z_m, z). \quad (6)$$

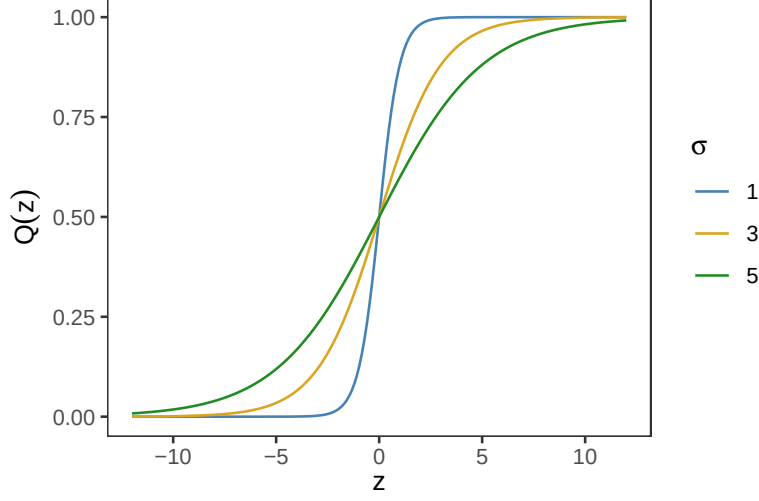


Figure 1: Shape of the sigmoid curve $Q(z)$ (Eqn 3) when varying the transition width σ (colors). The width σ is proportional to the distance over which the function transitions from values near zero to values near one.

To see if we can get evolutionary **branching**, we need to find a point where the **fitness gradient is zero** but its **curvature is positive**. We find the singular points by setting the fitness gradient to zero and finding the value of z that satisfies this condition:

$$\left. \frac{\partial G(z_m, z)}{\partial z_m} \right|_{z_m=z} = \frac{\partial}{\partial z_m} \left[b(z_m) - 2b(z)a(z_m, z) \right]_{z_m=z} = b'(z) - 2b(z) \left. \frac{\partial a(z_m, z)}{\partial z_m} \right|_{z_m=z}, \quad (7)$$

where $b'(z)$ is the derivative of b evaluated at z . Using Eqn 4:

$$\left. \frac{\partial a(z_m, z)}{\partial z_m} \right|_{z_m=z} = \frac{1}{2\sigma \cosh^2\left(\frac{z_m-z}{\sigma}\right)} \Big|_{z_m=z} = \frac{1}{2\sigma} \quad (8)$$

(because $\cosh(0) = 1$), so

$$\left. \frac{\partial G(z_m, z)}{\partial z_m} \right|_{z_m=z} = b'(z) - \frac{b(z)}{\sigma}. \quad (9)$$

This **gradient is zero** whenever

$$\sigma b'(z) = b(z). \quad (10)$$

To get the **curvature**, we compute the **second derivative** of the invasion fitness:

$$\left. \frac{\partial^2 G(z_m, z)}{\partial z_m^2} \right|_{z_m=z} = \frac{\partial^2}{\partial z_m^2} \left[b(z_m) - 2b(z)a(z_m, z) \right]_{z_m=z} = \left[b''(z_m) + \frac{\tanh\left(\frac{z_m-z}{\sigma}\right)}{\sigma^2 \cosh\left(\frac{z_m-z}{\sigma}\right)} \right]_{z_m=z}, \quad (11)$$

and since $\tanh(0) = 0$, this **reduces to the simple**

$$\left. \frac{\partial^2 G(z_m, z)}{\partial z_m^2} \right|_{z_m=z} = b''(z). \quad (12)$$

The curvature needs to exceed zero for evolutionary branching, so $b''(z) > 0$ is required. For example, functions that are convex everywhere do allow for branching, whereas those that are concave everywhere do not.

As an example, we can analyze the case of $b(z) = z^c$ for $z \geq 0$ (and $b(z) = 0$ for $z < 0$), where $c > 0$ is a constant exponent. Using Eqn 10 to find the singular trait value(s):

$$\sigma c z^{c-1} = z^c \implies z = c\sigma \quad (13)$$

(or $z = 0$, but then $b(0) = 0$ and the population cannot grow). In turn, the $b''(z) > 0$ condition reads

$$c(c-1)z^{c-2} > 0. \quad (14)$$

For $0 < c < 1$, this condition is violated and there are no evolutionary branching points. For $c > 1$, the inequality is satisfied and the convergence stable $z = c\sigma$ is also an evolutionary branching point. Figure 2 shows an example, with $c = 2$.

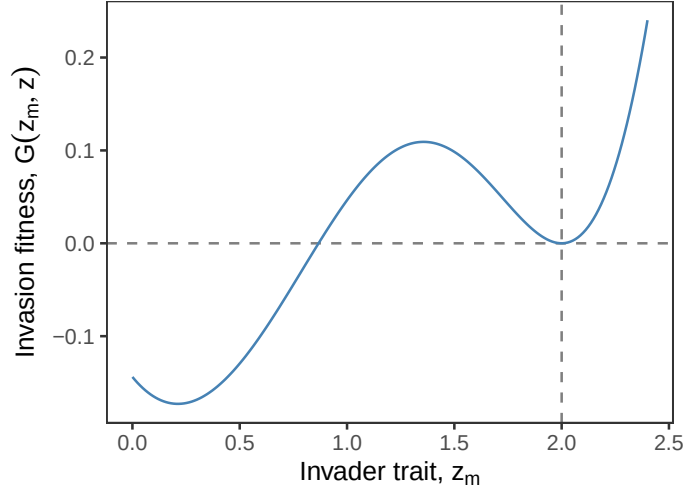


Figure 2: Invasion growth rate (blue curve) as a function of the invader trait z_m , when a resident species is sitting at its equilibrium density in the convergence stable point at $z = 2$ (highlighted by the vertical dashed line). The curvature of the fitness function around this point is clearly positive. The dashed horizontal line is drawn at zero growth. Parameters are $b(z) = z^2$, with $a(z_m, z)$ given by Eqn 4 and $\sigma = 1$.

For a more realistic example, we can look at an intrinsic growth function which initially increases in an accelerating manner but then eventually decelerates and flattens out. Indeed, a function such as $b(z) = z^2$ is not biologically realistic, because there are never any diminishing returns on investing more into the growth trait. Implementing such diminishing returns can be done via $Q(z)$ in Eqn 3:

$$b(z) = Q\left(\frac{z - z_0}{\eta}\right) - Q\left(-\frac{z_0}{\eta}\right). \quad (15)$$

The second term on the right hand side ensures $b(0) = 0$ and $b(z < 0) < 0$; z_0 is the curve's inflection point, and η is the transition width. For $0 < z < z_0$ we have $b''(z) > 0$, but for $z > z_0$, $b''(z) < 0$ (Figure 3).

This function also admits evolutionary branching. The singular points are still given by Eqn 10, after substituting in Eqn 15:

$$\frac{\sigma}{2\eta \cosh^2((z - z_0)/\eta)} = \frac{1}{2} \left[\tanh\left(\frac{z - z_0}{\eta}\right) - \tanh\left(-\frac{z_0}{\eta}\right) \right]. \quad (16)$$

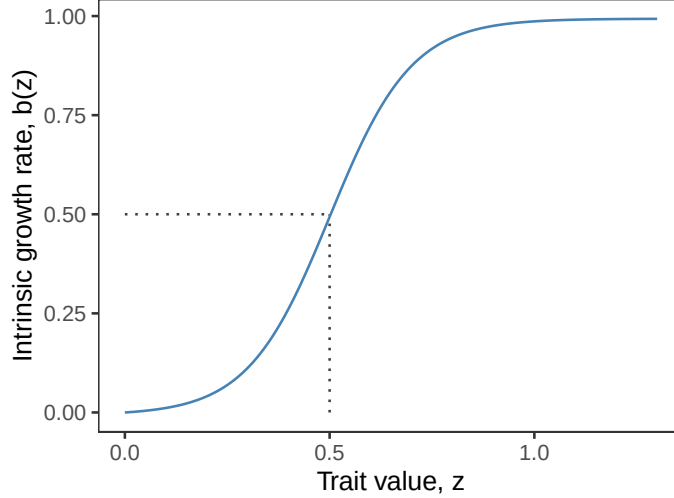


Figure 3: The intrinsic growth rate function of Eqn 15, with $z_0 = 0.5$ and $\eta = 0.2$. The dotted lines show the inflection point.

This transcendental equation can be solved via e.g. numerical root solving. For example, choosing $z_0 = 0.5$, $\eta = 0.2$, and $\sigma = 0.15$, its solution is $z \approx 0.436$. To determine if this admits evolutionary branching, we check if Eqn 12 is satisfied at this point. Since the inflection point of $b(z)$ is at $z_0 = 0.5$, we have that $b''(z) > 0$ for $0 < z < 0.5$. As $z \approx 0.436$ does fall in this range, this value of z is indeed an evolutionary branching point.

Eco-evolutionary simulations

Let us implement an eco-evolutionary simulation of the model. For exploratory purposes, we use Eqn 15 for the intrinsic rates, with $\eta = 0.2$ and $z_0 = 0.5$. We still use Eqn 4 for the competition kernel, with $\sigma = 0.13$. We follow a “poor man’s adaptive dynamics” approach:

1. Start with a single phenotype somewhere;
2. Integrate Eqn 1 until equilibrium;
3. Introduce a new mutant at a low density and with a trait value that is a slight change from the resident’s trait;
4. Repeat from 2, with the difference that the mutant trait value in 3 can be a slight change of any one persisting phenotype (randomly picked) from the current community.

We now implement the simulation. The system starts with a single phenotype at $z = 0.3$. The number of mutation events is 10 000, with mutations normally distributed around a randomly chosen community member in each iteration with standard deviation 0.001. This random selection is weighted by the population densities: a larger population has a higher chance to be selected for producing the next mutant. The newly-generated phenotype’s invasion density is always set to 0.001. After every integration of the ecological dynamics, species with population density below the threshold of 0.003 are removed. Given these simulation hyperparameters, the results are in Figure 4.

We can also compute the community matrix (Jacobian evaluated at equilibrium). First, let us look at the final population densities and trait values:

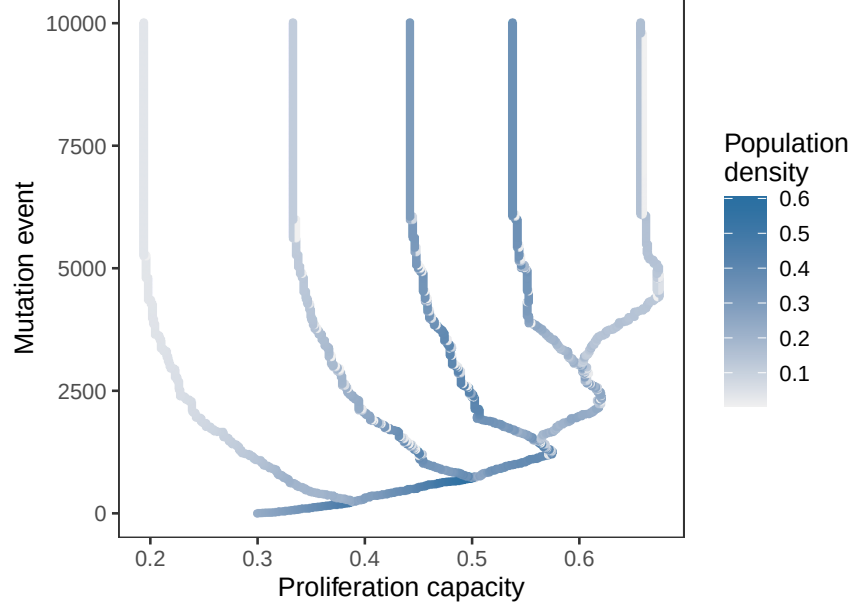


Figure 4: Eco-evolutionary outcome of Eqn 1 after 10 000 mutation events. A single initial phenotype at $z = 0.3$ diversifies into a stable community of five clones.

clone	trait	density
1	0.1944	0.0354
2	0.3331	0.1149
3	0.4419	0.2975
4	0.5374	0.3510
5	0.6565	0.1608

We can now compute and visualize the corresponding community matrix. The result is in Figure 5.

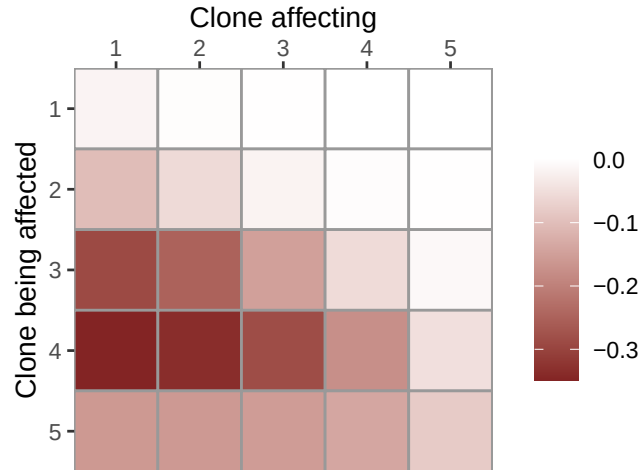


Figure 5: Community matrix of the final eco-evolutionary state in Figure 4. The resulting matrix is nearly lower triangular, signaling a hierarchical structure.

The matrix in Figure 5 is also stable, having eigenvalues that are all real and negative:

eigenvalues
-0.0060
-0.0141
-0.0328
-0.0852
-0.3418

Furthermore, its inverse exhibits the typical checkerboard pattern of alternating positive and negative entries that is typical of hierarchical competition-colonization models (Figure 6). The (i, j) th entry of this inverse measures the long-term effect on species i when species j experiences a sustained press perturbation.

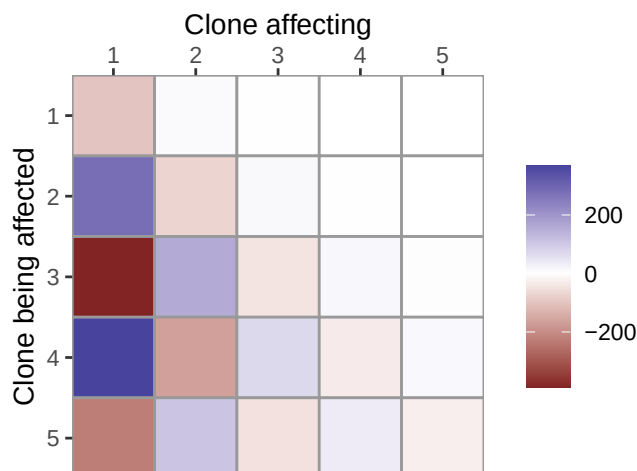


Figure 6: Inverse of the matrix in Figure 5.

References

Adler, F. R., and J. Mosquera. 2000. Is space necessary? Interference competition and limits to biodiversity. *Ecology* 81:3226–3232.